IGCSE 0607 Practice Questions — E2.7 — Combined Questions & Markscheme

Sequences

Total Questions: 6 — **Total Marks:** 33 Marks

Table of Contents & Quick Navigation

- **E2.7** Sequences
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

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1 Here are the first four terms of a sequence.

5 12 19 26

(a) Write down the next term in the sequence.

..... **[1]**

(b) Find an expression for the nth term of the sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

2 Here are the first four terms of a sequence.

5 11 17 23

(a) Find the next term in this sequence.

..... **[1]**

(b) Find an expression for the nth term of this sequence.

..... **[2]**

(c) Explain why 150 is not a term in this sequence.

..... **[1]**

3 The first five terms of a sequence, T_n, are given below.

$$4 \qquad 14 \qquad 34 \qquad 68 \qquad 120$$

(a) By finding the first, second and third differences, show that the nth term of the sequence is a cubic expression in n.

[2]

(b) Find an expression for T_n, the nth term of the sequence.

$T_n =$ ..... **[5]**

4 **(a)** The nth term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ... , 2nd term = ... , 3rd term = ...
[2]

(b) The first four terms of another sequence are

2 9 20 35

Find an expression for the nth term of this sequence.

..... **[3]**

5 **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the nth term of this sequence.

..... **[4]**

(b) An exponential sequence has nth term $P_n = 5 \times 3^{n-1}$.
The kth term of this sequence is 3645.
Work out the value of k.

$k =$ ..... **[2]**

(c) The nth term of sequence T is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that T_n approaches as n becomes very large.

..... **[1]**

6 The first four terms of a sequence are

$$-1 \quad 2 \quad 9 \quad 20$$

(a) Find the nth term of this sequence.

..... **[3]**

(b) The nth term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

1 **(a)** 33 **[B1]**

(b) $7n + k$ for any integer k **[M1]**

$7n - 2$ **[A1]**

(c) $7n - 2 = 150$

$7\text{n} = 152 \mathit{which does not give an integer value for n.} [B1]$

[Back to Question] *[Total Marks: 4]*

2 **(a)** 29 **[B1]**

(b) $6n + c$ or finding common difference of 6 **[M1]**

$6n - 1$ **[A1]**

(c) Sets $6n - 1 = 150 \implies 6n = 151$ and states that 151 is not divisible by 6 (or n is not an integer) **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** First differences: $10, 20, 34, 52$ **[M1]**

Second differences: $10, 14, 18$
Third differences: $4, 4$
Since the third difference is a non-zero constant, the sequence is cubic. **[A1]**

(b) Uses $6a = 4 \implies a = \frac{2}{3}$ **[M1]**

Uses $12a + 2b = 10 \implies 12(\frac{2}{3}) + 2b = 10 \implies 8 + 2b = 10 \implies b = 1$ **[M1]**

Uses $7a + 3b + c = 10 \implies 7(\frac{2}{3}) + 3(1) + c = 10 \implies c = 10 - 3 - \frac{14}{3} = \frac{7}{3}$ **[M1]**

Uses $a + b + c + d = 4 \implies \frac{2}{3} + 1 + \frac{7}{3} + d = 4 \implies 4 + d = 4 \implies d = 0$ **[M1]**

$\frac{2}{3}n^3 + n^2 + \frac{7}{3}n$ **[A1]**

[Back to Question] *[Total Marks: 7]*

4 **(a)** One correct term evaluated **[M1]**

$3, 3, 5$ **[A1]**

(b) Second differences are 4 **[M1]**

Identifies $2n^2$ as part of the expression **[M1]**

$2n^2 + n - 1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Finds third differences are 6 (first diffs: $9, 25, 47, 75$; second diffs: $16, 22, 28$; third diffs: 6) **[M1]**

Identifies n^3 as the cubic term **[B1]**

Subtracts n^3 from the sequence to get a quadratic sequence $(-2, 0, 6, 16, 30)$ and finds its second differences are 4 to give $2n^2$ **[M1]**

$n^3 + 2n^2 - 4n$ **[A1]**

(b) $5 \times 3^{k-1} = 3645$

$3^{k-1} = 729$ [M1]

7 **[A1]**

(c) $\frac{2}{3}$ or 0.667 **[B1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** Finds first differences: $3, 7, 11$ and second differences: $4, 4$ **[M1]**

Deduces the term involves $2n^2$ **[M1]**

$2n^2 - 3$ **[A1]**

(b) $3^n - 2 \approx 1000 \implies 3^n \approx 1002$ **[M1]**

Uses logarithms or trial and improvement: $n = 6 \implies 3^6 - 2 = 729 - 2 = 727$; $n = 7 \implies 3^7 - 2 = 2187 - 2 = 2185$ **[M1]**

727 **[A1]**

[Back to Question] *[Total Marks: 6]*